Experimental results: core evidence

Common setup. Four checkpoint-bound BBB providers, 157 held-out molecules, one call per input, and $D \in \{0, 1, 4\}$. Each method has 30 schedules and 4,710 recorded calls. Controlled drift flips the selected fit-best provider's label after tick 78. All paired intervals resample scaffold clusters.

Table 1: Pre-call selection on BBB: natural outcomes and recovery under controlled correctness drift.

	Natural	Controlled drift		
Method	Success (%)	Success (%)	Gain (pp)	95% CI
Random	83.76	—	—	—
Best Fixed	88.15	—	—	—
Frozen	**89.81**	73.38	reference	—
MIRA-Relative	**89.81**	—	—	—
Online SGD	89.21	84.03	+10.66	[+6.05, +15.45]
Greedy LinUCB	89.72	81.15	+7.77	[+5.08, +10.72]
BayesianRouter	89.53	80.28	+6.90	[+3.50, +10.41]
MIRA-Relay	**89.81**	**87.56**	+14.18	[+9.61, +19.03]
Retrospective Best Fixed†	90.45	—	—	—

Gain is paired against Frozen. † Evaluator-only fixed provider. Bold marks the best deployable point estimate, including ties. —: not in the recorded roster.

On the natural menu, MIRA-Relay's paired gain is 0.00 pp (95% CI $[-1.24, +1.40]$). It rescues and harms 82 decisions each. Under the specified drift, it rescues 780 and harms 112 decisions.

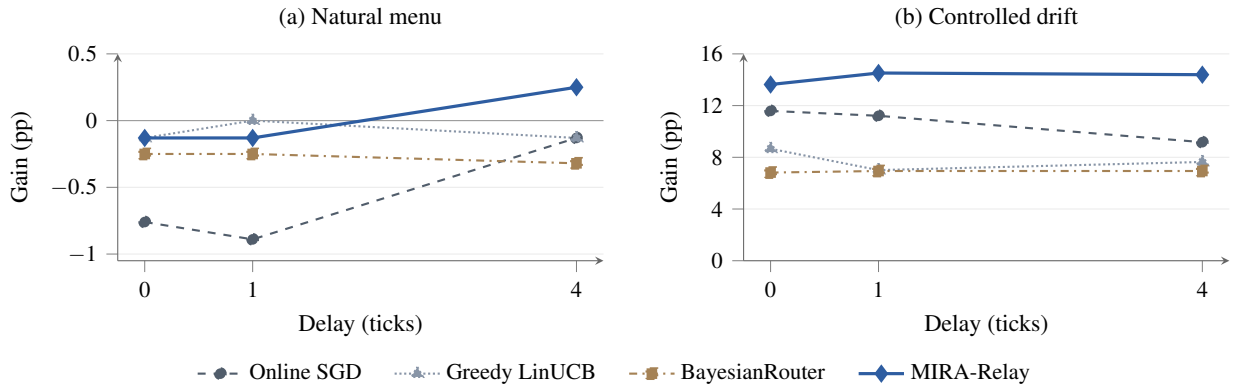


Figure 1: Delay sensitivity. Paired gains over Frozen at the recorded delays; points are descriptive, with no uncertainty shown.

Table 2: Source-aligned attribution with matched cold start and release clock.

Contrast	Natural Δ [95% CI]	Controlled Δ [95% CI]
Local − Frozen	+0.10 [−1.08, +1.42]	+14.40 [+9.79, +19.23]
Permutation − Frozen	+0.41 [−0.47, +1.32]	−1.27 [−3.66, +0.95]
Local − Permutation	−0.30 [−1.42, +0.76]	+15.67 [+10.65, +20.74]

Units: pp. Equal-weight scaffold-mean contrasts over 124 clusters; 90 replay streams per condition. This estimand differs from the pooled-call point estimates in Table 1.

Development diagnostics and deployment cost

Development setup. The diagnostics use 722 molecules and five scaffold folds. Offline objectives use paired seeds; the update factorial uses three delays and 2,166 decisions per cell.

Table 3: Cold-start accuracy on BBB development. The per-molecule oracle is evaluator-only.

Selector	Accuracy (%)
Best Fixed	79.22
Raw Frozen	79.36
Corrected Frozen	**80.19**
BCE + auxiliary CLS	78.53 / 78.95
BT + auxiliary CLS	78.81 / 78.25
Per-molecule oracle†	85.60

Paired training seeds 0/1 are separated by '/'; one value denotes a deterministic result. † Uses unselected outputs.

Table 4: Score scale and update rule on BBB development. Accuracy in %.

Score	Update	Natural	Drift
Raw	Frozen	79.36	69.11
	Marginal	79.18	77.29
	Query	78.49	75.95
Corrected	Frozen	80.19	63.30
	Marginal	79.82	78.30
	Query	80.33	78.67

Compare update rules within each scale; score correction also changes exposure to the intervention.

The development panel has 471 all-correct, 147 mixed-correctness, and 104 all-wrong molecules. Corrected Frozen gets 579/722 right; the oracle gets 618/722, leaving 39 recoverable selection errors. Neither offline objective closes this gap in the tested recipe.

Table 5: CPU policy overhead (μs; lower is better).

	Natural menu		Controlled
Method	Selection	Update	Update
Random	67.14	5.44	—
Best Fixed	5.82	0.68	—
Frozen	15.89	1.31	1.50
MIRA-Relative	42.02	73.30	—
Online SGD	14.08	19.37	20.02
Greedy LinUCB	128.63	90.90	92.32
BayesianRouter	133.65	88.96	92.55
MIRA-Relay	27.21	2.81	3.46

Mean of 30 stream-level p95 timings; macOS arm64. Excludes provider inference and I/O. All methods use one call per input. Live learner memory remains unmeasured.

Table 6: Post-call refusal on the reused BBB holdout (secondary analysis).

Policy	Coverage (%)	Accepted accuracy (%)	Utility
Always accept	100.00	88.15	−0.1285
Confidence refuse	84.46	**94.57**	**−0.0947**
Always refuse	0.00	—	−0.2600
Same-coverage random†	84.46	88.69	−0.1444

† Evaluator-side conditional null. $U = -(W + 0.25R + 0.01C)/N$: W accepted errors, R refusals, C calls, N records. All policies call once; bold marks the best deployable point estimate.

This secondary analysis reuses the 157-molecule holdout: five policies fitted to development folds yield 785 records per policy. Confidence minus random is +5.88 pp accepted accuracy (95% CI $[+2.77, +9.29]$); utility versus always accept improves by +0.0338 (95% CI $[+0.0088, +0.0609]$). It does not estimate pre-call online gain.

Experiment map. E1 → Table 1 and Figure 1; E2 → Table 2; E3 → Table 3; E4 → Table 4; E5 → Table 5; E6 → Table 6. Protocols and source hashes: companion README.